

ARCH 594/508: AI Research Seminar/Studio—AI in Architecture
Spring 2026

MACHINE LEARNING

Supervised Learning: Classification

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REVIEW

In our previous sessions:

Machine Learning

Introduction on ML Types and Applications

Supervised Learning

- Prediction
- Regression
- Feature Selection
- Model Evaluation

Computational Hands-On Activity

- Regression

AGENDA FOR TODAY'S CLASS

Machine Learning Fundamentals

Supervised Learning

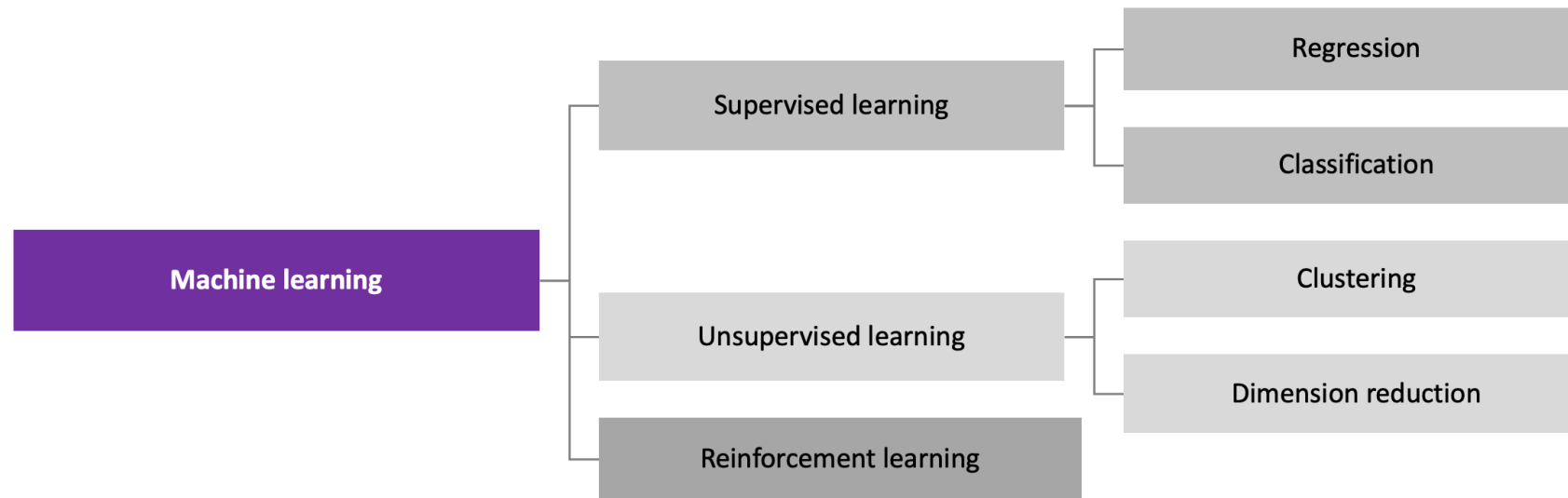
- Classification
- Model Selection & Generalization
- Model Evaluation
- Confusion Matrix
- Cross Validation

Computational Hands-On Activity

- Classification

REVIEW

AN INTRODUCTION TO MACHINE LEARNING TYPES AND APPLICATIONS



REVIEW

SUPERVISED LEARNING

Definition: Supervised learning is a type of machine learning where algorithms learn patterns and associations between input features and their corresponding output or target values using **labeled training data**.

Types:

- **Regression:** Model predicts **continuous values**.
- **Classification:** Model predicts **discrete class labels**.

REVIEW

UNSUPERVISED LEARNING

Definition: Unsupervised Learning uses **unlabeled data** for exploration and pattern discovery.

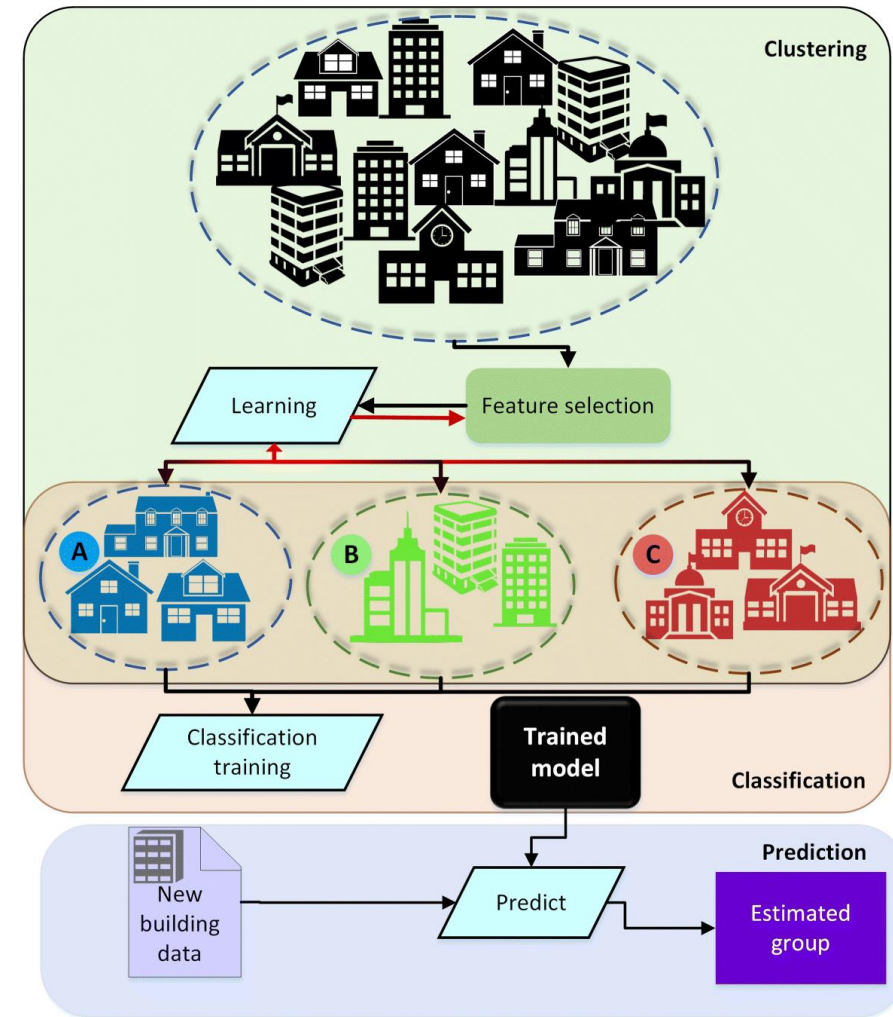
Key Components:

- **Unlabeled data:** input data patterns without known outcomes.
- **Learning patterns:** Algorithm focuses on unveiling hidden structures, clusters, or relationships within the data.

Techniques in Unsupervised Learning:

- **Clustering:** Grouping similar data points.
- **Dimensionality Reduction:** Reducing complexity by extracting essential features.

REVIEW EXAMPLES



<https://viejournal.springeropen.com/articles/10.1186/s40327-018-0064-7>

SUPERVISED LEARNING

REVIEW

PREDICTION IN MACHINE LEARNING



REVIEW

PREDICTION IN MACHINE LEARNING



Algorithm: Take feature values as input and predict the outcome.



REVIEW

REGRESSION

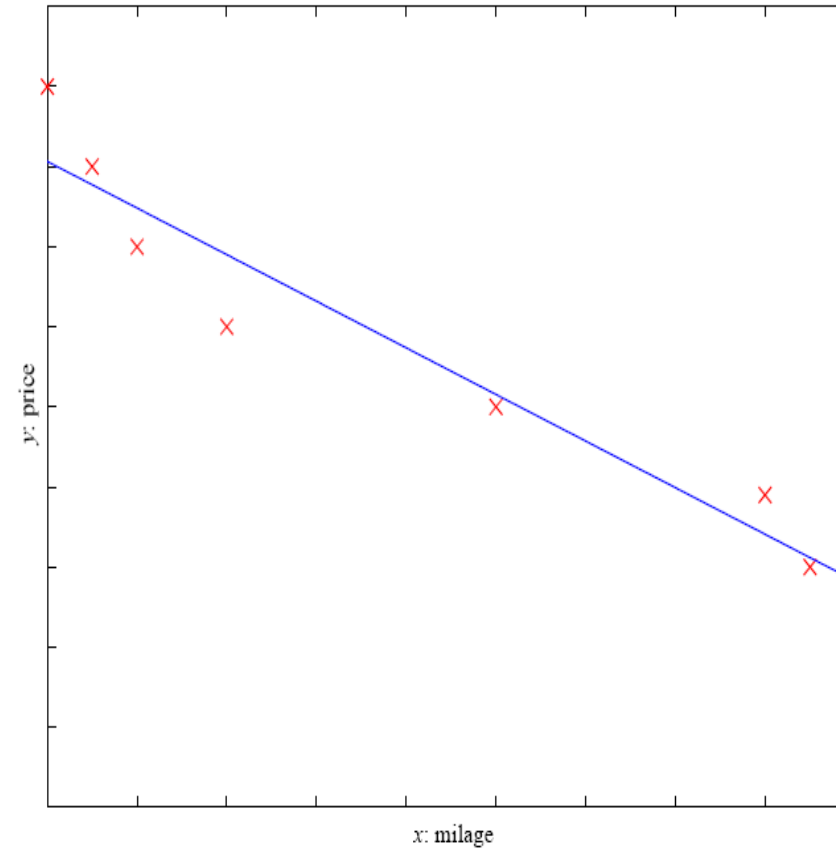
- A statistical technique that models the relationship between a dependent variable and independent variable(s) to **predict continuous numerical outcomes**.
- Example: Price of a used car
- x : car attributes

y : price

$$\hat{y} = F(x | \theta)$$

$F()$ model,

θ parameters



CLASSIFICATION

Prediction when the outcome is Categorical

- Categorical outcome tasks termed as "classification."
- Classification is the process of categorizing data into distinct classes or categories based on their features or attributes.

DECISION RULES FOR CATEGORICAL OUTCOMES

- Models output a **decision rule**; it determines which of the 'k' classes to predict based on the give predictors.

$$f_k(x_1, x_2, \dots, x_p)$$

- Example: Decision rule based on (f_1) of (x_1 through x_p) compared to a pre-defined constant (C) for binary data.

For **binary** categorical outcomes:

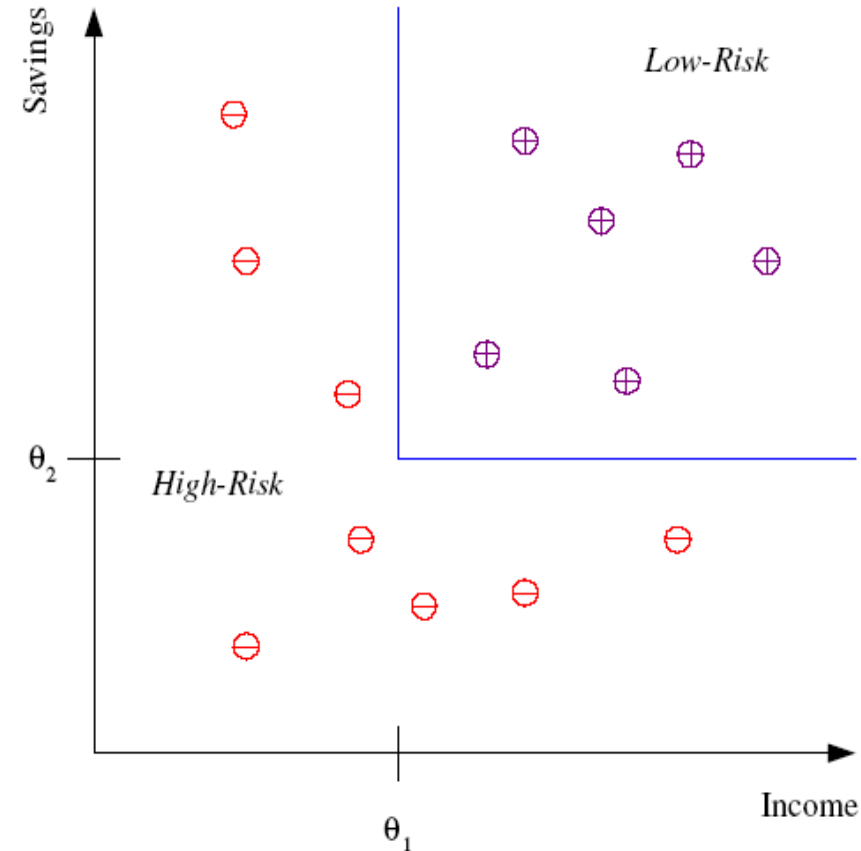
If $f_1(x_1, x_2, \dots, x_p) > C$ predict category 1.

If $f_1(x_1, x_2, \dots, x_p) \leq C$ predict category 0.

Since the outcome is categorical the result is either right or wrong.

EXAMPLE CLASSIFICATION

- Example: Credit scoring
- Differentiating between **low-risk** and **high-risk** customers from their *income* and *savings*



IF *income* $> \theta_1$ AND *savings* $> \theta_2$
THEN **low-risk** ELSE **high-risk**

LEARNING A CLASS FROM EXAMPLES

Prediction:

Is car x a family car?

- Class C of a “family car”
- Output/Outcome:
 - Positive (+) and negative (−) examples
- Input(s)/Feature(s):
 - x_1 : price, x_2 : engine power

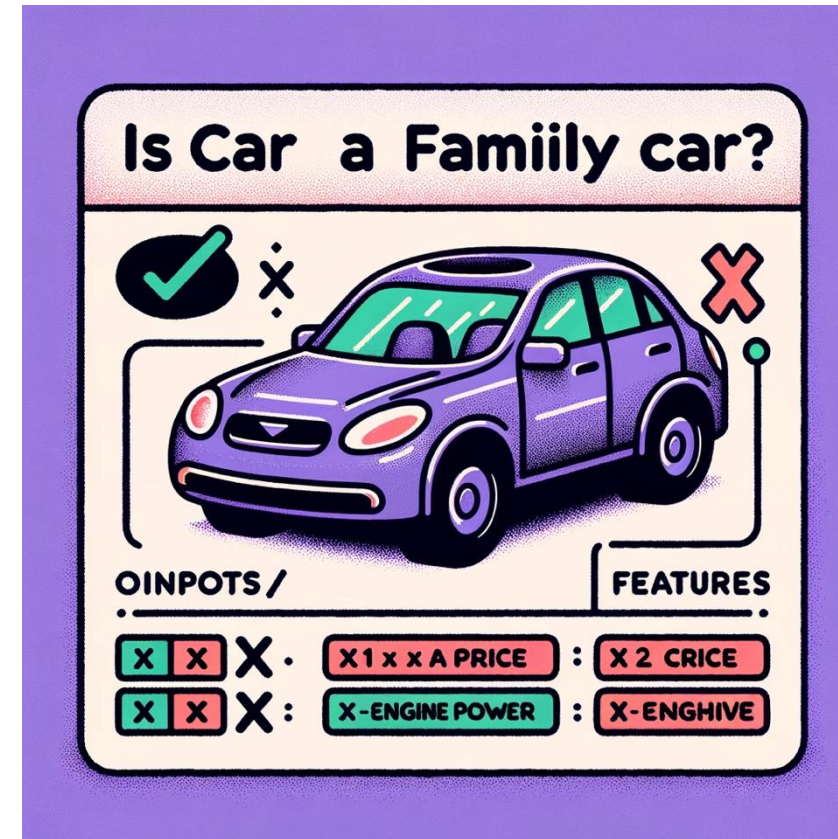
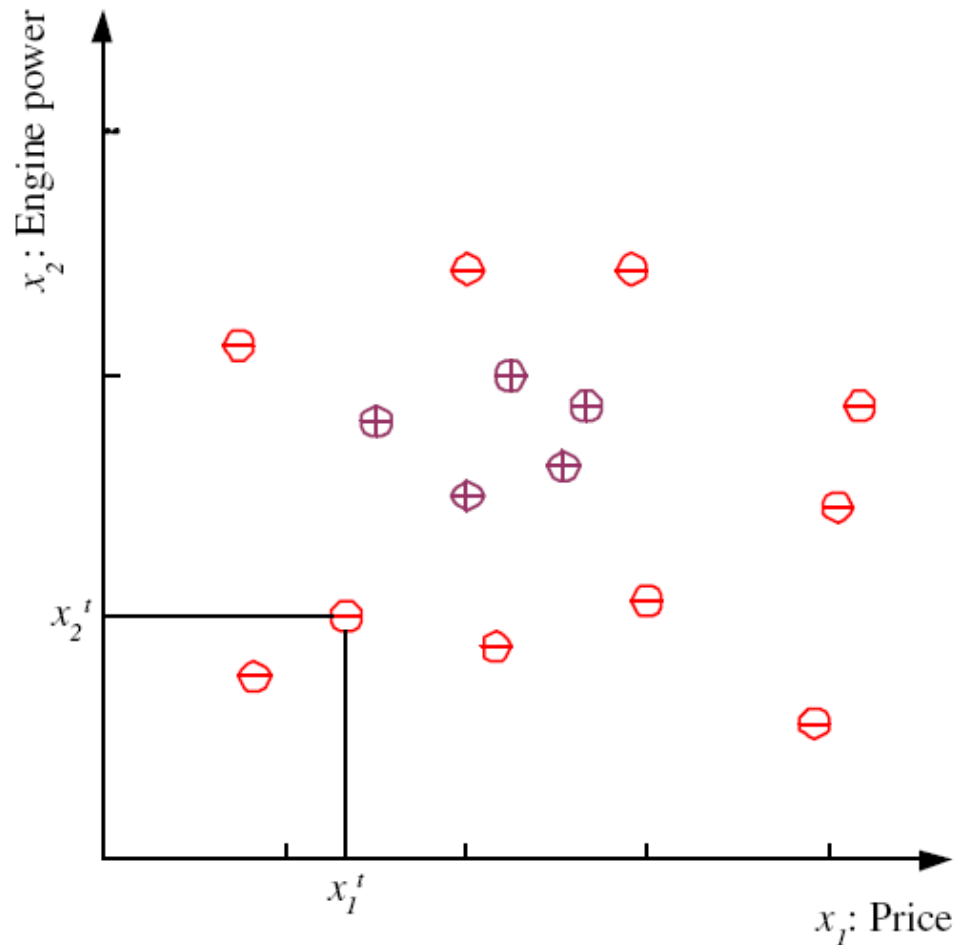


Image Source: OpenAI. DALL-E.

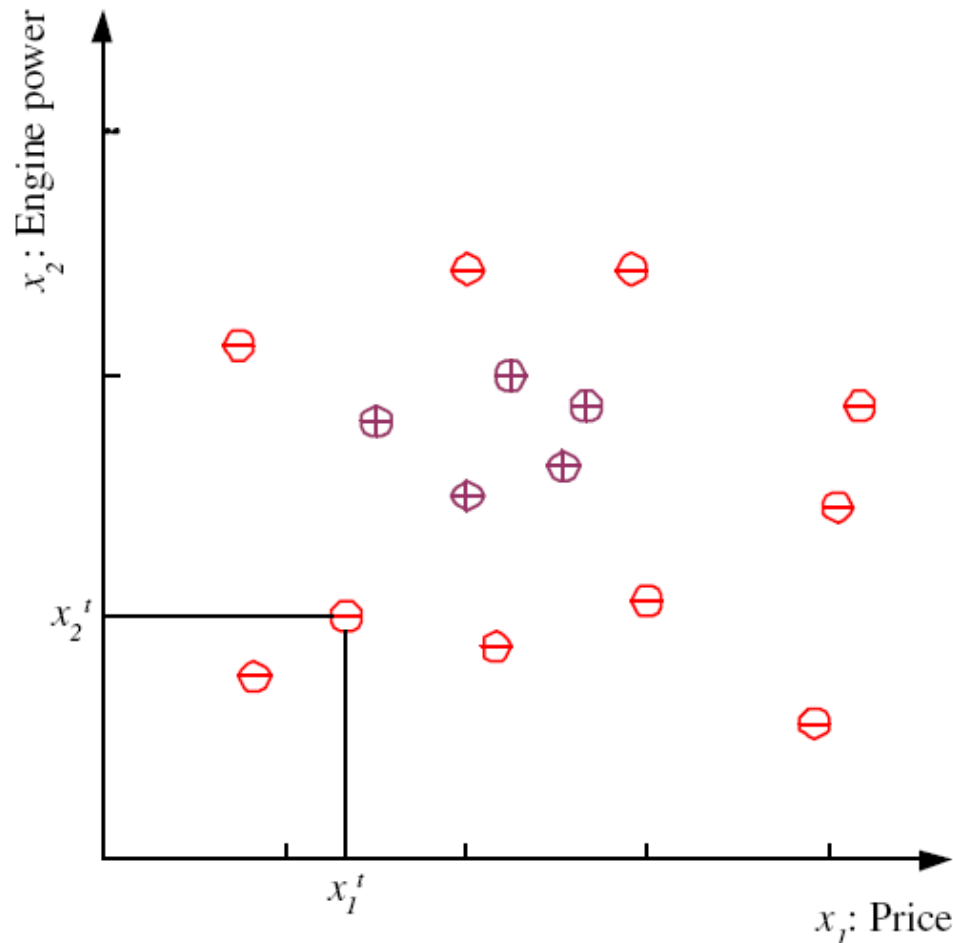
15

TRAINING SET $\mathcal{X}$



- Output/Outcome:
Positive (+) and negative ($-$) examples
- Input(s)/Feature(s):
 x_1 : price, x_2 : engine power

TRAINING SET $\mathcal{X}$



1

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

2

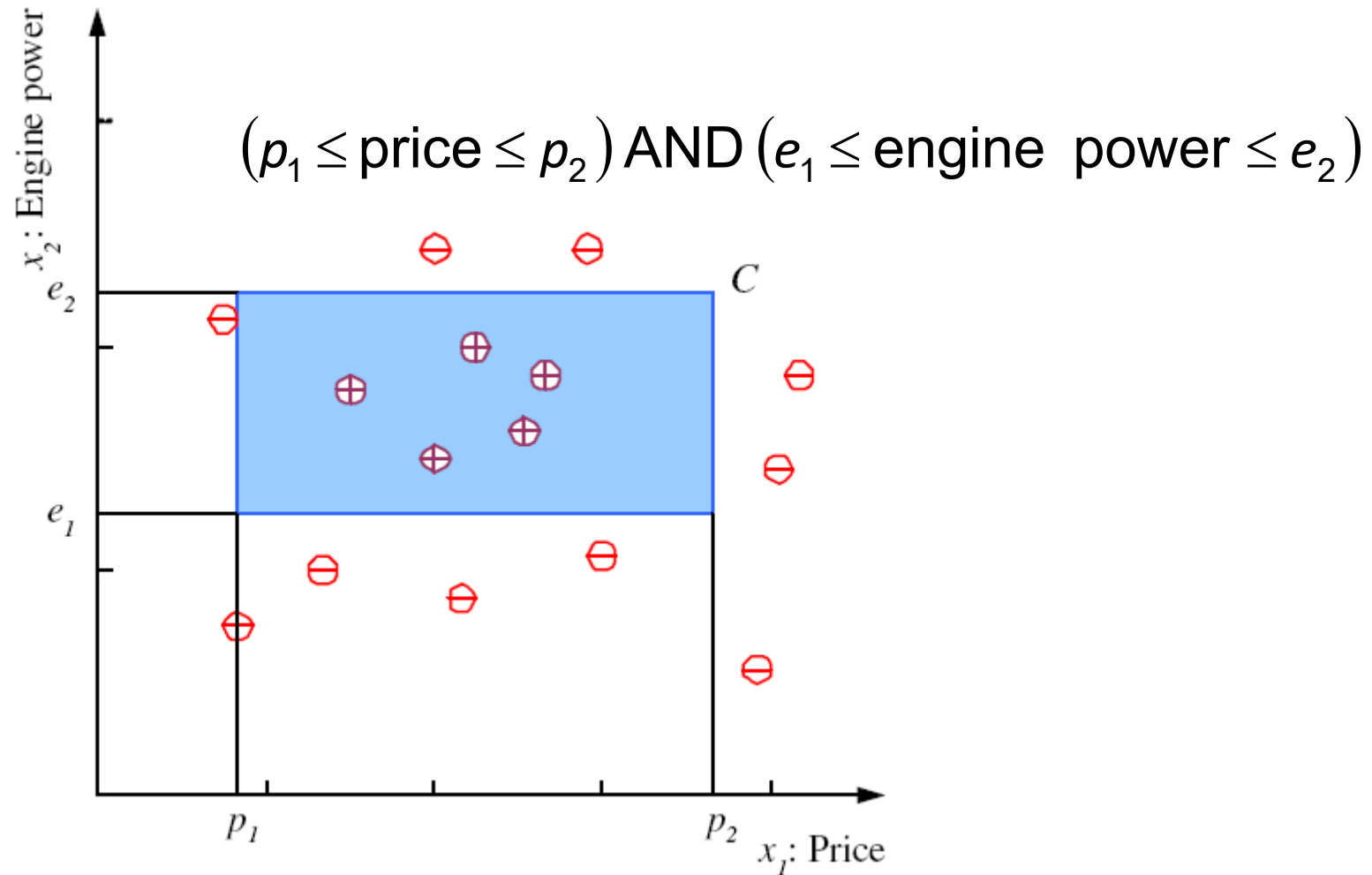
$$r = \begin{cases} 1 & \text{if } \mathbf{x} \text{ is positive} \\ 0 & \text{if } \mathbf{x} \text{ is negative} \end{cases}$$

3

$$\mathcal{X} = \{\mathbf{x}^t, r^t\}_{t=1}^N$$

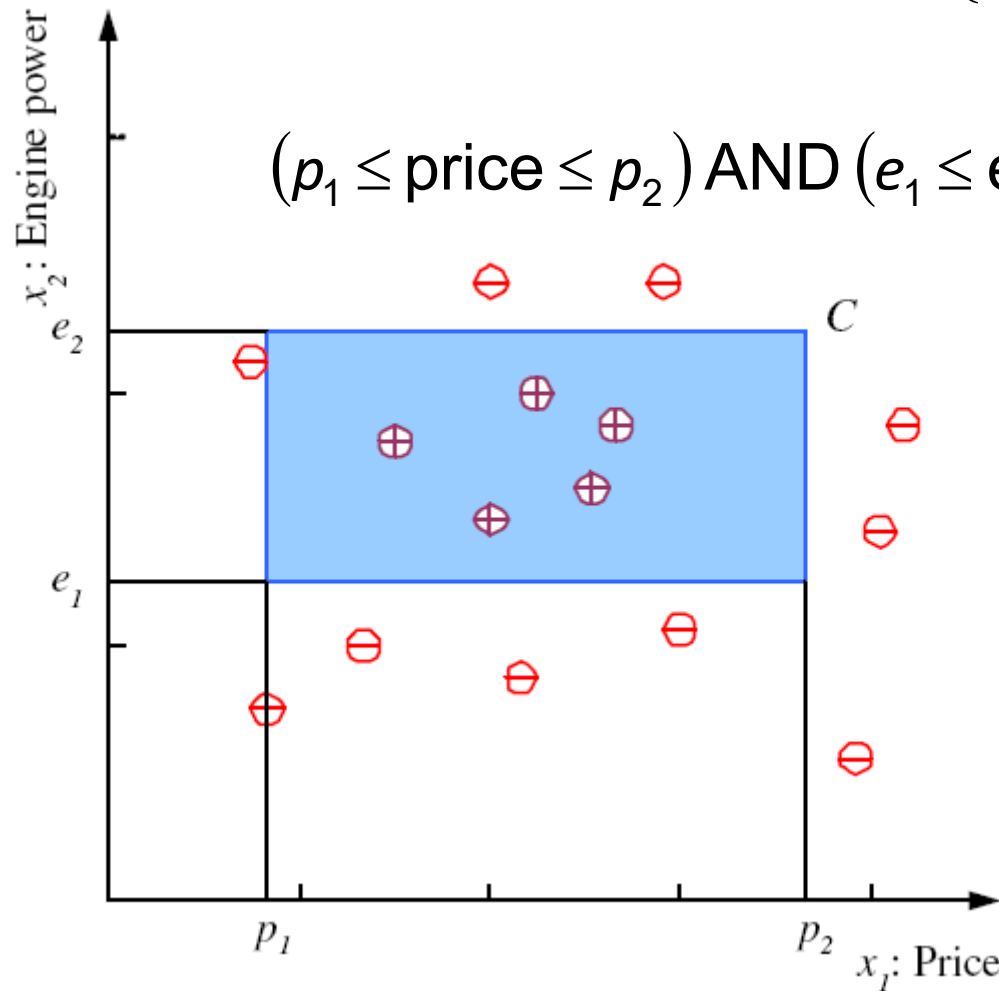
- $\mathcal{X}$: Represents the entire training set, which contains N examples.
- $\mathbf{x}$: A vector of two dimensions, representing each car with attributes x_1 (price) and x_2 (engine power).
- r : The label for each car, where $r = 1$ if it's a positive example (family car) and $r = 0$ if it's a negative example (not a family car).
- t : An index representing different examples in the set; it's not related to time or any specific order.
- N : The total number of examples in the training set.

CLASS C

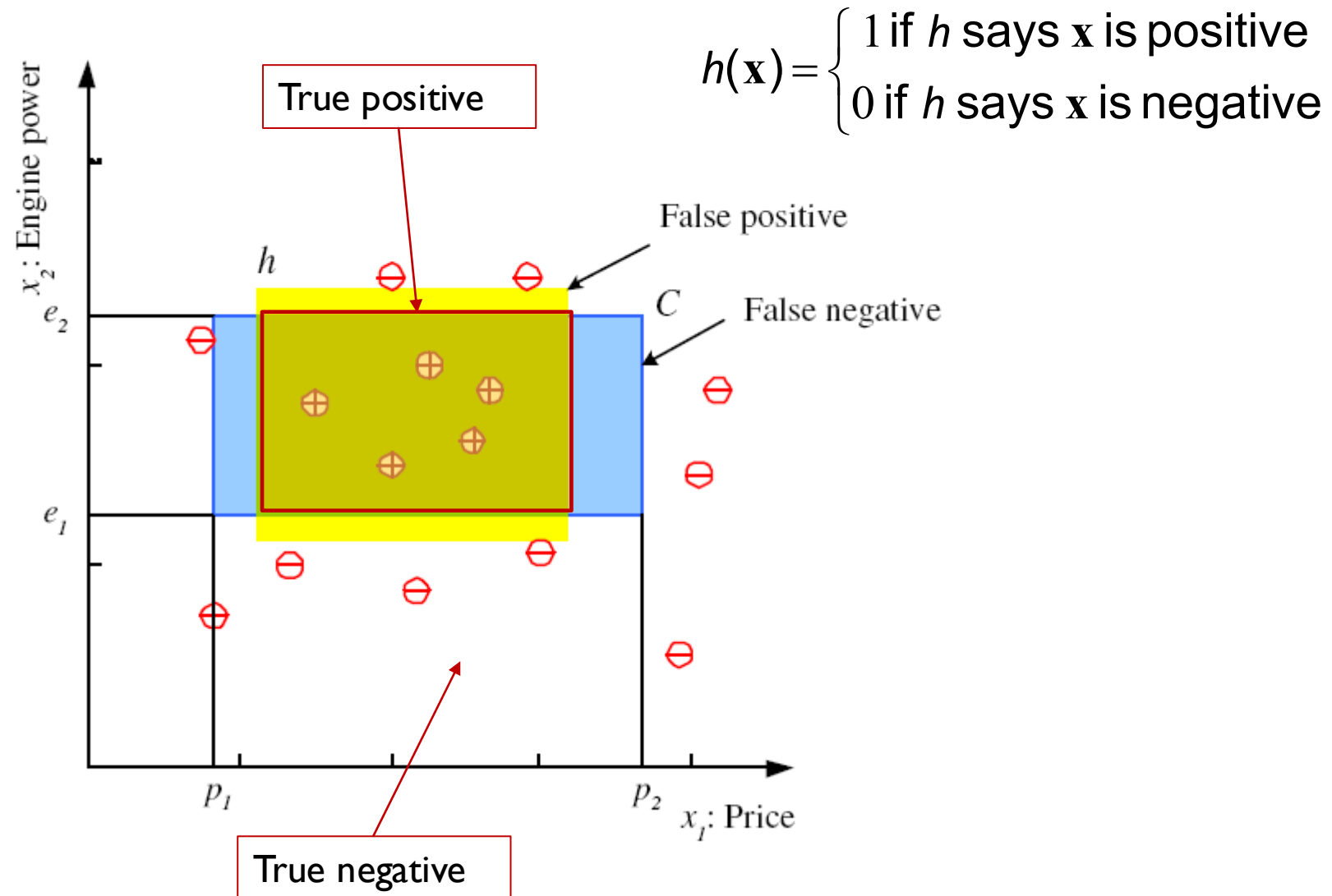


HYPOTHESIS CLASS $\mathcal{H}$

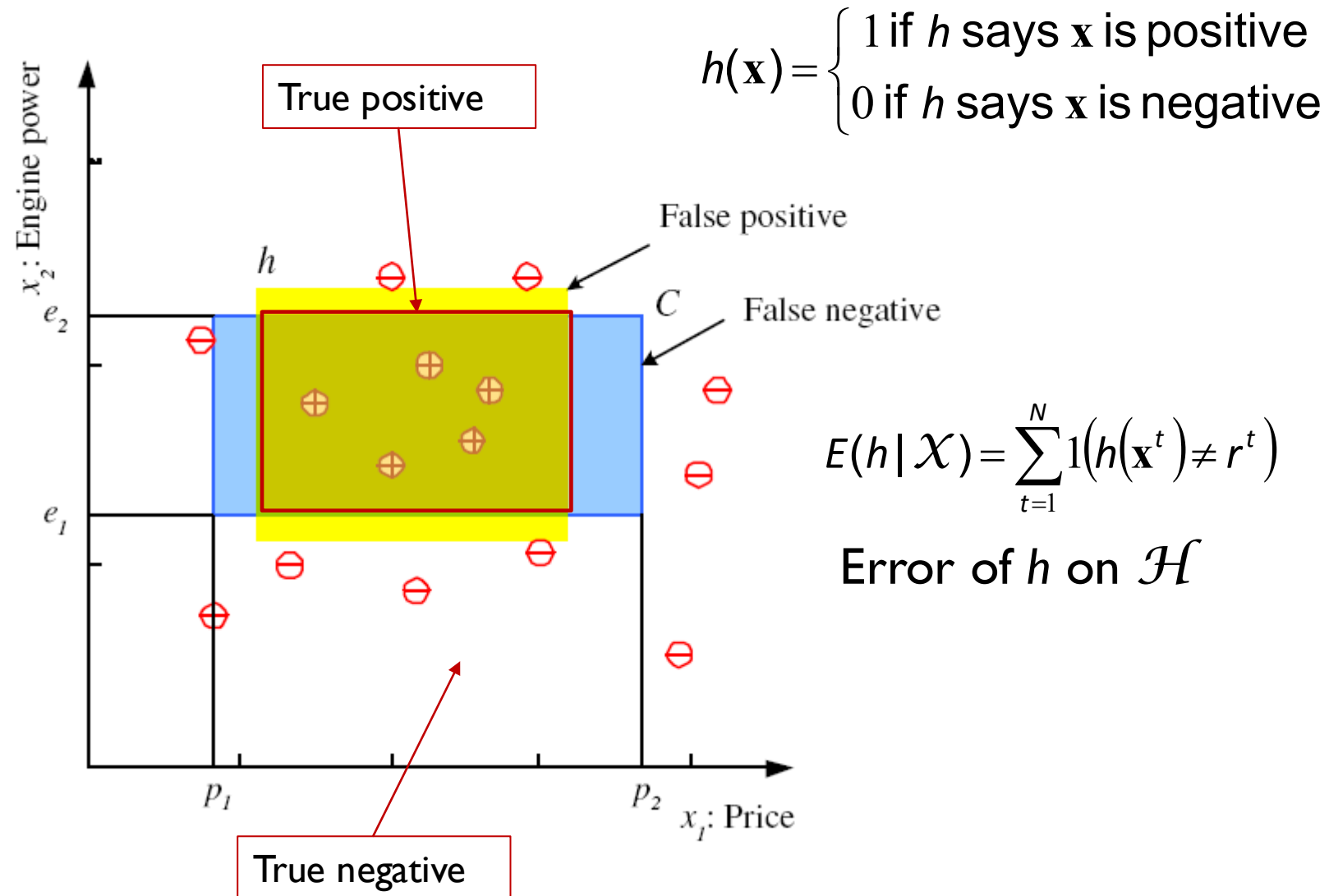
$$h(\mathbf{x}) = \begin{cases} 1 & \text{if } h \text{ says } \mathbf{x} \text{ is positive} \\ 0 & \text{if } h \text{ says } \mathbf{x} \text{ is negative} \end{cases}$$



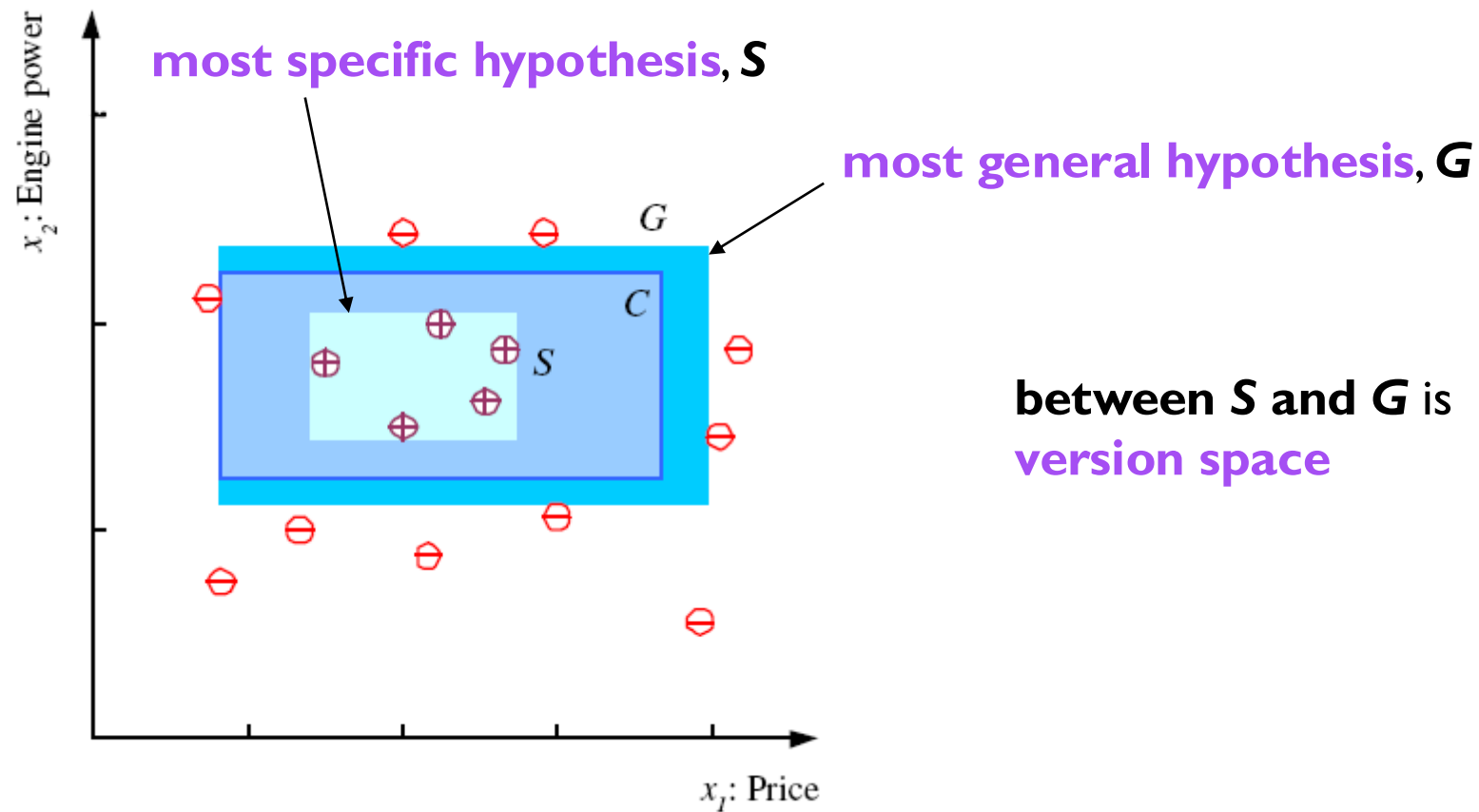
HYPOTHESIS CLASS $\mathcal{H}$



HYPOTHESIS CLASS $\mathcal{H}$

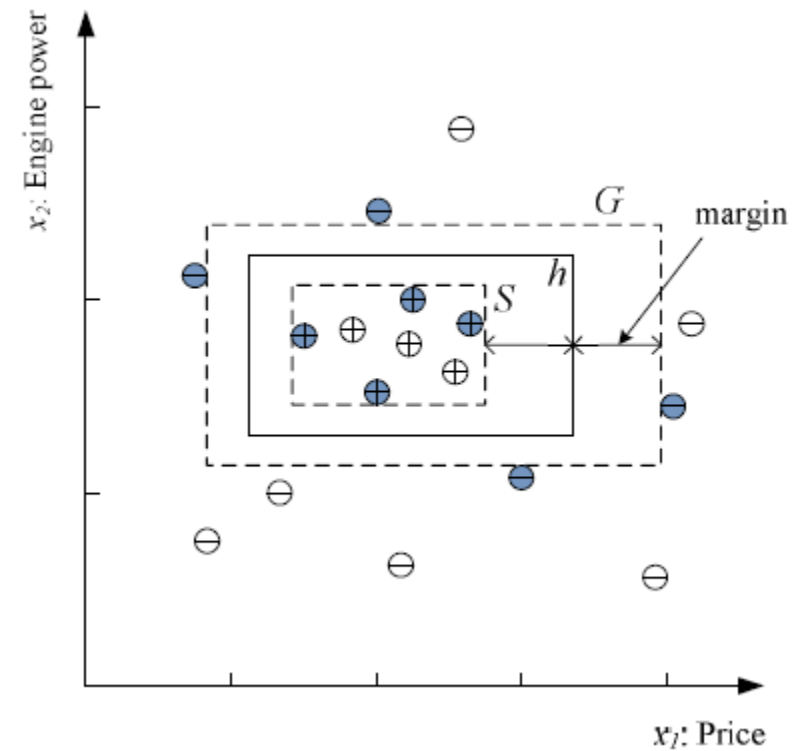


S, G, AND THE VERSION SPACE



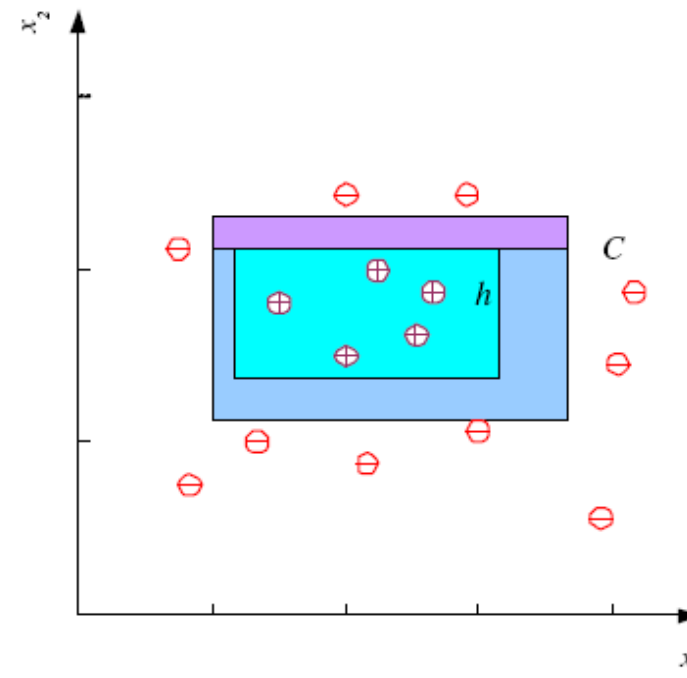
MARGIN

- With X available, we can choose S , G , or any h from the version space as our hypothesis.
- Opting for h between S and G increases the margin, the **distance to the nearest instances**.
- In cases of costly errors, instances between S and G may lead to uncertainty.



PROBABLY APPROXIMATELY CORRECT (PAC) LEARNING

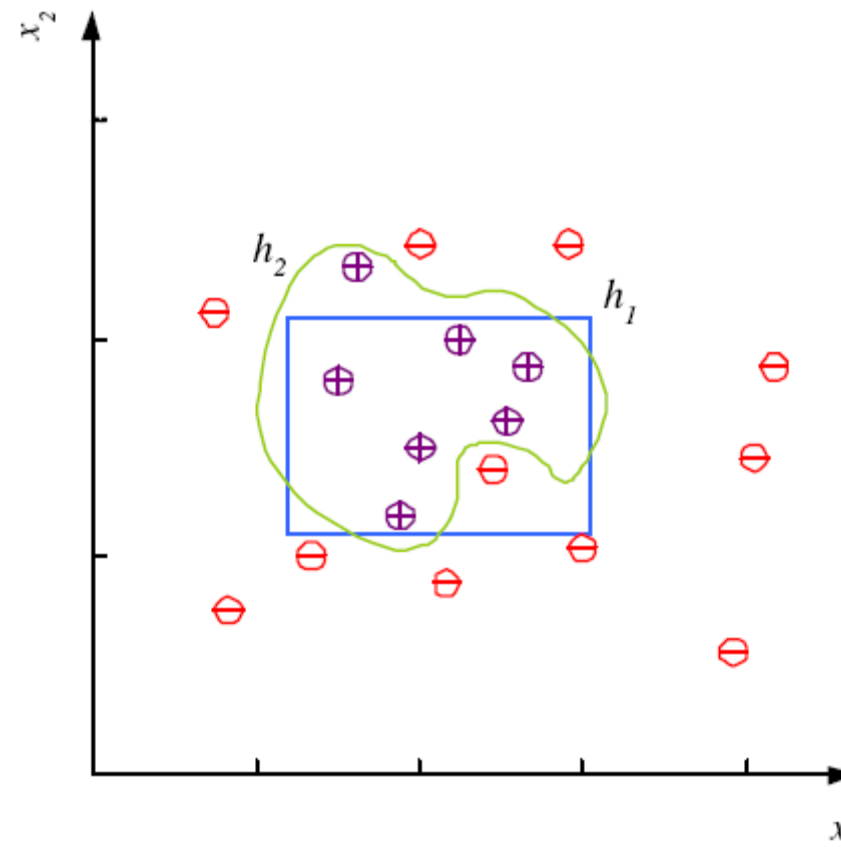
- Goal is to **find a hypothesis that approximately fits the data.**
- This involves determining the number of examples needed to ensure the hypothesis remains close to the actual pattern.



NOISE AND MODEL COMPLEXITY

Noise in data refers to **unwanted irregularities** that can make learning a class more challenging and may prevent a simple model from achieving perfect accuracy

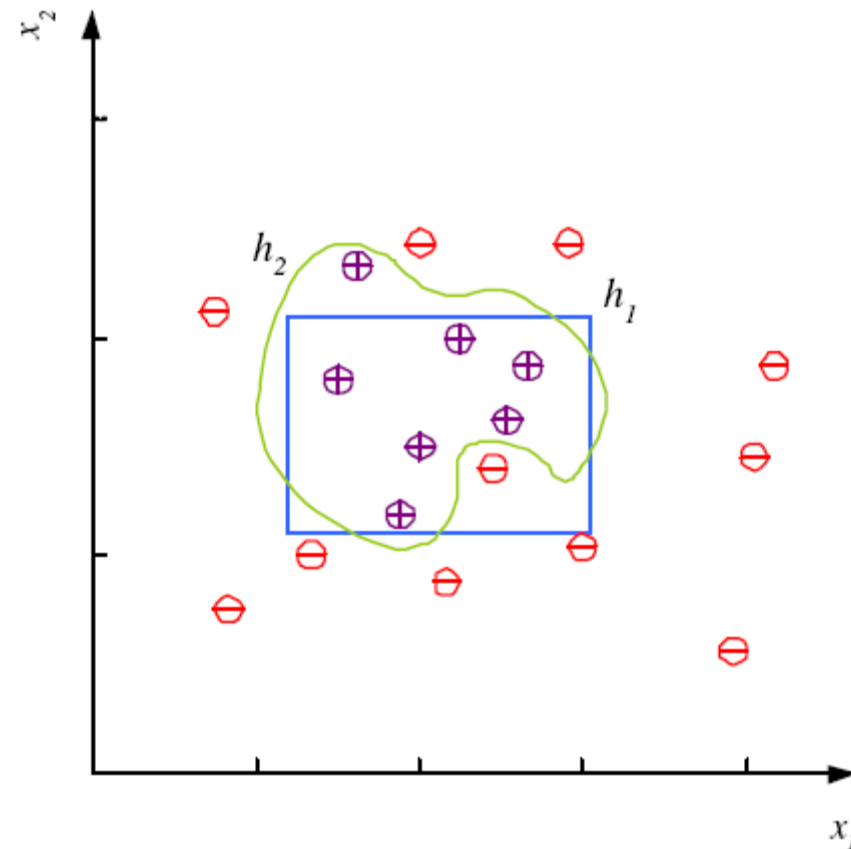
When there is noise, there is not a simple boundary between the positive and negative instances.



NOISE AND MODEL COMPLEXITY

Noise can arise from various sources:

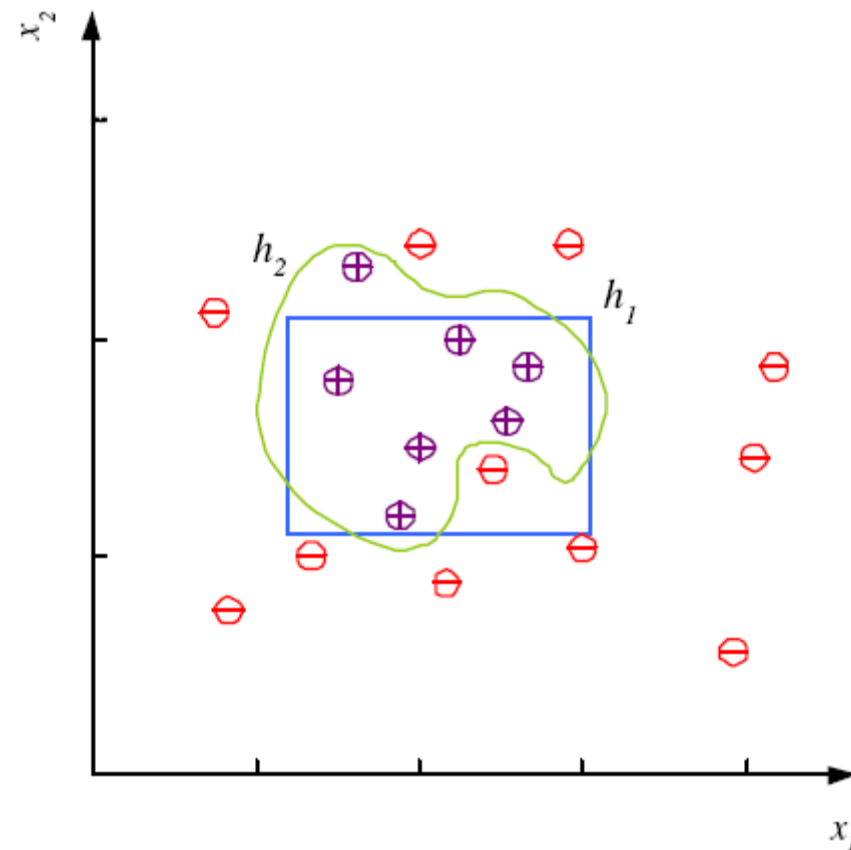
- Imperfections in Recording Data
- Labeling Errors
- Additional Unaccounted Attributes



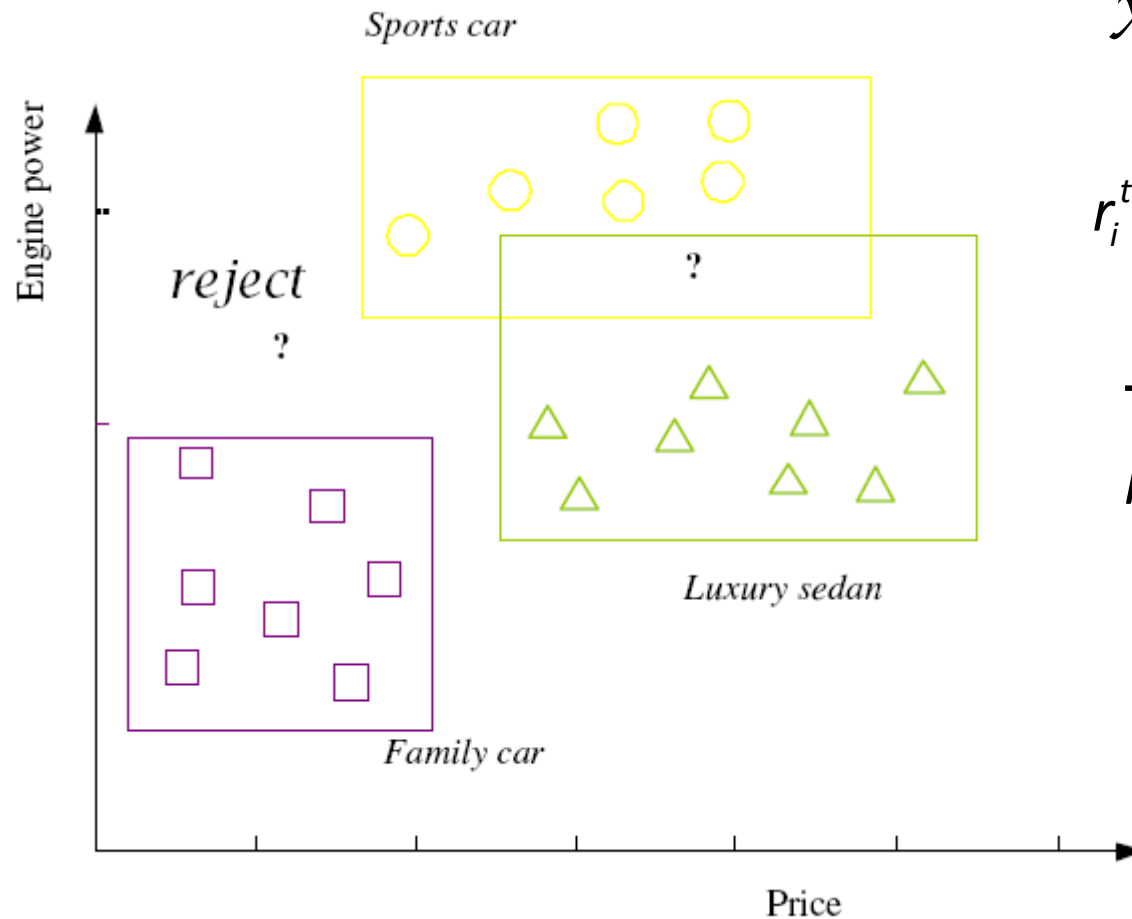
NOISE AND MODEL COMPLEXITY

Using a simple model like a rectangle makes sense for several reasons:

- **Simpler to use** (lower computational complexity)
- **Easier to train** (lower space complexity)
- **Easier to explain** (more interpretable)
- **Generalizes better to new data** (lower variance)



MULTIPLE CLASSES, $C_i \ i=1,...,K$



$$\mathcal{X} = \{\mathbf{x}^t, r^t\}_{t=1}^N$$

$$r_i^t = \begin{cases} 1 & \text{if } \mathbf{x}^t \in C_i \\ 0 & \text{if } \mathbf{x}^t \in C_j, j \neq i \end{cases}$$

Train hypotheses

$h_i(\mathbf{x}), i = 1, \dots, K:$

$$h_i(\mathbf{x}^t) = \begin{cases} 1 & \text{if } \mathbf{x}^t \in C_i \\ 0 & \text{if } \mathbf{x}^t \in C_j, j \neq i \end{cases}$$

MODEL SELECTION & GENERALIZATION

- **Model selection:** Model selection is the process of choosing the best model from a set of candidate models for a specific task. It balances the trade-off between model complexity and performance, aiming to find a model that generalizes well to unseen data.

MODEL SELECTION & GENERALIZATION

- **Generalization:** How well a model performs on new data
 - Generalization in machine learning refers to the ability of a model to make accurate predictions on new, unseen data based on its training on existing data.

UNDERFITTING AND OVERFITTING

- **Underfitting:** $\mathcal{H}$ less complex than $\mathcal{C}$ or f
 - Underfitting occurs in machine learning when the hypothesis (H) is too simple to capture the underlying patterns in the training data, resulting in poor performance on both the training and new data.
 - (e.g., fitting a line to polynomial data).
- **Overfitting:** $\mathcal{H}$ more complex than $\mathcal{C}$ or f
 - Overfitting happens in machine learning when H is excessively complex and fits the training data too closely, resulting in poor performance on new, unseen data as it fails to generalize effectively.
 - (e.g., using complex functions with noisy data)

TRIPLE TRADE-OFF

There is a trade-off between three factors:

1. **Complexity of H , $c(H)$,**
2. **Training set size, N ,**
3. **Generalization error, E , on new data**

- As training data increases, generalization error decreases.
- With an increase in model complexity (class H), generalization error initially decreases but then begins to rise.

DIMENSIONS OF A SUPERVISED LEARNER MODEL

- **Loss Function:** Measures how far off the model's predictions are from the actual outcomes.
 - **Note:** It's the core feedback mechanism that helps the model learn and improve.

$$E(\theta | \mathcal{X}) = \sum L(r^t, g(\mathbf{x}^t | \theta))$$

- **Optimization Procedure:** The method used to minimize the loss function by adjusting the model's parameters.
 - **Note:** Optimization algorithms (like gradient descent) are used to find the best model configuration.

$$\theta^* = \arg \min_{\theta} E(\theta | \mathcal{X})$$

CONFUSION MATRIX

A Confusion Matrix is a visualization tool/table for **assessing the accuracy of classification** models by **comparing its predicted labels with the actual labels**.

		Predicted Label	
		+	-
True Label	+	True Positive (TP)	False Negative (FN)
	-	False Positive (FP)	True Negative (TN)

CONFUSION MATRIX

- The confusion matrix:
 - True Positives (TP): Correctly predicted positive instances.
 - True Negatives (TN): Correctly predicted negative instances.
 - False Positives (FP): Incorrectly predicted positive instances (Type I error).
 - False Negatives (FN): Incorrectly predicted negative instances (Type II error).
- The confusion matrix helps in understanding different types of classification errors.

		Predicted Label	
		+	-
True Label	+	True Positive (TP)	False Negative (FN)
	-	False Positive (FP)	True Negative (TN)

CONFUSION MATRIX

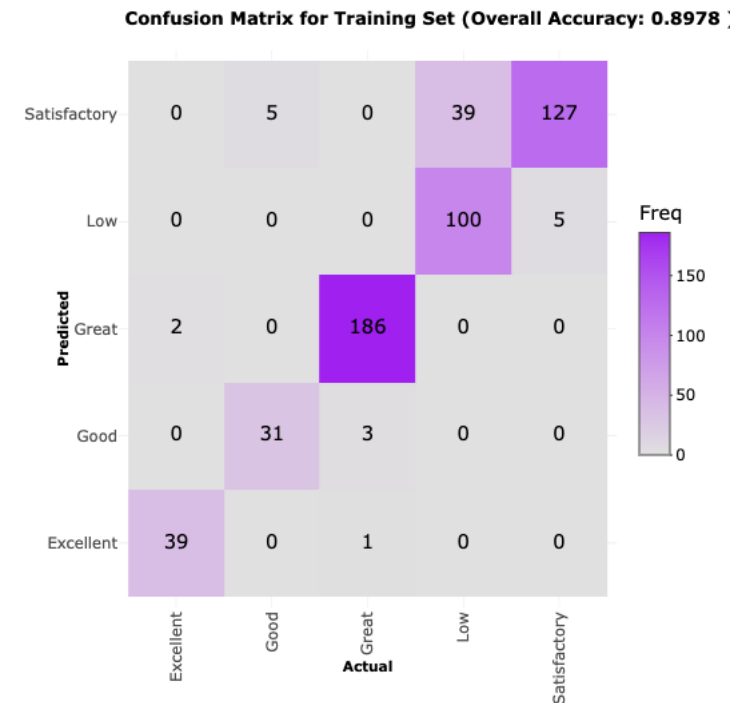
Example: Energy Efficiency Classification

■ Structure:

- Rows represent predictions made by the model,
- Columns indicate the true categories.

■ Insights:

- Helps identify not just the count of correct and incorrect predictions, but the nature of the errors ;
- Enables detailed performance analysis for each class, aiding in fine-tuning of the classifier.



IMPLICATIONS OF FALSE POSITIVES VS. FALSE NEGATIVES

- **False Positives (FP):** In the context of building energy efficiency classification, a false positive means classifying a building as energy efficient when it is not. This could lead to poor policy decisions or missed opportunities for improvement.
- **False Negatives (FN):** A false negative means failing to identify an energy-efficient building. This could result in missed incentives or failure to recognize best practices.
- Understanding the implications of these errors is crucial in various contexts:
 - In building performance for making informed decisions in building design, energy policy and management.
 - How about in medical?

MODEL EVALUATION METRICS

- Different evaluation metrics that can be used to assess the performance of a classification model:
 - **Accuracy**
 - **Precision**
 - **Recall**
 - **F1 Score**
- These metrics provide different perspectives on how well a model is performing and can help identify trade-offs in performance.

ACCURACY, PRECISION, RECALL, AND F1-SCORE

- **Accuracy:** The ratio of correctly predicted instances to the total instances. It is useful when classes are balanced.
$$\text{Accuracy} = (\text{Number of Correct Predictions}) / (\text{Total Number of Predictions})$$
- **Precision:** The ratio of correctly predicted positive observations to the total predicted positives. It answers the question: 'How many of the predicted positives are actually positive?'
$$\text{Precision} = (\text{True Positives}) / (\text{True Positives} + \text{False Positives})$$
- **Recall (Sensitivity):** The ratio of correctly predicted positive observations to all observations in the actual class. It answers: 'How many actual positives are captured by the model?'
$$\text{Recall} = (\text{True Positives}) / (\text{True Positives} + \text{False Negative})$$
- **F1-Score:** The harmonic mean of precision and recall. It is useful when you need a balance between precision and recall.
$$\text{F1 Score} = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$$

CROSS-VALIDATION

- To estimate generalization error, we need data unseen during training. We split the data as

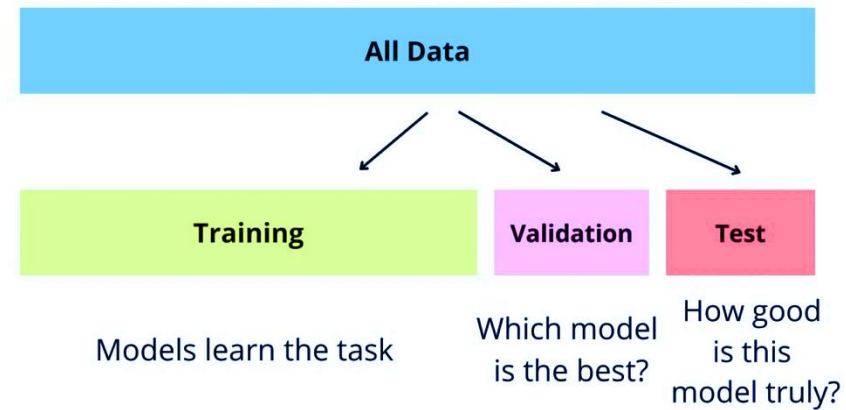


Image Source: galaxyinferno.com

K-FOLD CROSS-VALIDATION

- where the **dataset is divided into 'K' equally-sized subsets** or "folds."
 - For **example**, if K is set to 5, the data is split into five parts.
 - **Training and Testing:** The cross-validation process then repeats 'K' times.
 - **Performance Evaluation:** After each iteration, the model's performance is measured on the testing fold.
 - **Average Performance:** Once all 'K' iterations are complete, the performance metrics obtained from each fold are averaged to get a single performance score for the model.

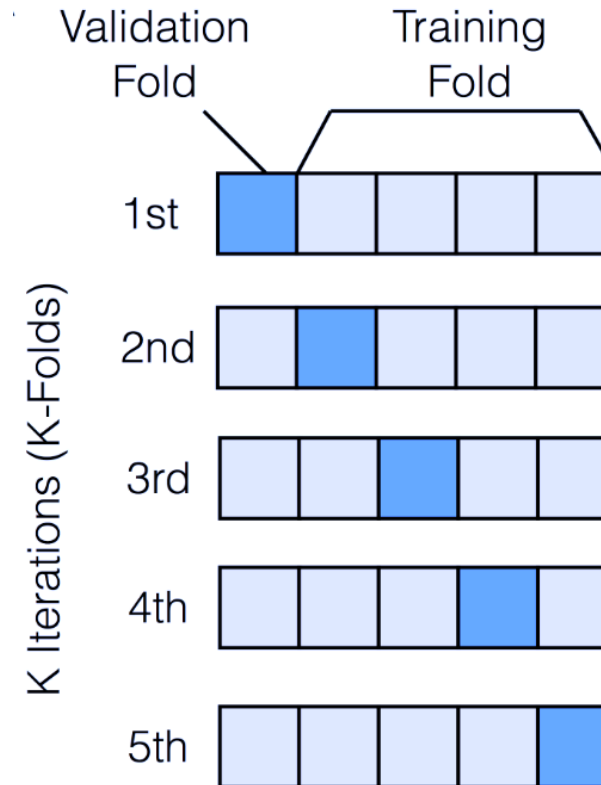


Image Source: medium.com

CONCLUSION

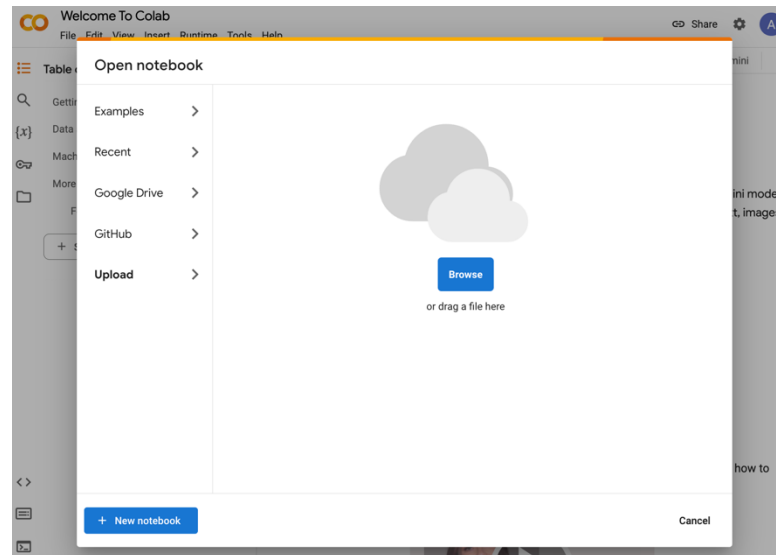
Machine Learning Fundamentals

Supervised Learning

- Classification
- Model Selection & Generalization
- Confusion Matrix
- Model Evaluation
- Cross validation

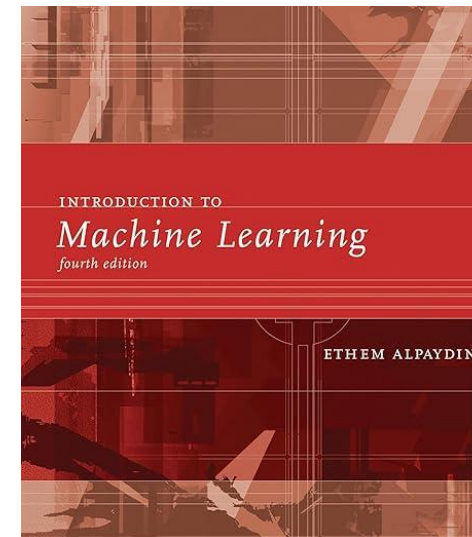
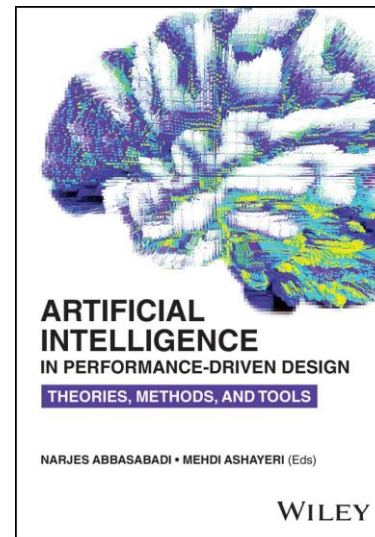
LAB ACTIVITY: CLASSIFICATION

1. Open up a web browser. Google “google colab” click on the first entry. You may also access the website by following <https://colab.research.google.com/>
2. Go to Canvas and download the file called “Classification - Categorizing Building Energy Efficiency Levels in Seattle.ipynb”



MAIN TEXTBOOKS

- Abbasabadi, N., Ashayeri, M., (Eds). (2024). Artificial Intelligence in Performance-driven Design: Theories, Methods and Tools, Wiley.
- Alpaydin, E., (2020). Introduction to Machine Learning, 4th Edition, The MIT Press



READINGS

- Alpaydin, E., (2020). Introduction to Machine Learning. 4th Edition, The MIT Press. – **Chapter 1 / What is Machine Learning?**
– **Chapter 2 / Supervised Learning**

NEXT CLASSE

- Unsupervised Learning
 - Clustering
 - Lab Activities

QUESTIONS?